

PosoKanei API Reference

Undocumented backend of posokanei.gov.gr, reconstructed from the open-source posokanei-mcp client code and verified with real responses where possible.

Base URL: `https://api.posokanei.gov.gr`

Important: this is not an official, published API. It is the internal backend used by the PosoKanei website and apps. It can change or stop working at any time, and requests should be kept modest (the reference client waits at least 250 ms between requests and caches responses for 5 minutes).

How to read the status labels

Label	Meaning
CONFIRMED	The endpoint was called for real and the raw JSON was inspected (history_sample.json, competitors_sample.json, retailers_sample.json, stats_sample.json, categories_sample.json).
WORKING	The call is used successfully by the tracker scripts. Product-level fields are cross-checked against real product data, but the raw envelope was not dumped separately.
FROM CLIENT CODE ONLY	Parameters and shape come from reading the posokanei-mcp source. Not tested live.

Overview

Request requirements

The API sits behind bot protection. A bare script request (for example Python requests with its default User-Agent) can receive **403 Forbidden**. Sending browser-like headers resolved this in testing:

```
accept: application/json
content-type: application/json
user-agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
  (KHTML, like Gecko) Chrome/128.0.0.0 Safari/537.36
origin: https://posokanei.gov.gr
referer: https://posokanei.gov.gr/
accept-language: el-GR,el;q=0.9,en;q=0.8
```

No API key or login is used. If 403 responses return, the protection has probably changed.

Endpoint summary

#	Method	Path	Purpose	Status
1	POST	/products/search	Search products by text or ids	Working
2	GET	/products	List products in a category	Client code only
3	GET	/products/{id}	One product, prices, optional history	Confirmed
4	GET	/products/barcode/{barcode}	One product by barcode	Client code only
5	GET	/products/{id}/competitors	Similar products in the same category	Confirmed
6	GET	/meta/categories/tree	Category tree with VAT data	Confirmed
7	GET	/meta/retailers	Directory of supermarket chains	Confirmed
8	GET	/meta/stats	Dataset-wide counters	Confirmed
9	GET	/meta/categories	Flat category list (unused)	Untested

MCP tool to endpoint map

Tool	Endpoint(s)	Notes
search_products	1	Renames query to title, pageSize to page_size, etc.
get_product	3	
lookup_barcode	4	Barcode must be 8, 12 or 13 digits (validated client-side).
list_products_by_category	2	Defaults to sort_by unit_price.

Tool	Endpoint(s)	Notes
list_categories	6	maxDepth and maxItems are applied client-side after the response arrives.
list_retailers	7	Passed through unchanged.
get_price_history	3	Same endpoint as get_product with include_history=true.
compare_competitors	5	
compare_basket	1, 3, 4	No endpoint of its own. Client-side orchestration, up to 25 items, totals per retailer.
get_stats	8	

Common objects

Product object (ProductSummary)

Returned inside products[] (endpoints 1, 2), as the top-level object (3, 4) and inside competitors[] (5, reduced field set). The client types mark most fields optional, so always read them defensively (dict.get in Python).

Field	Type	Notes
id	string	Stable product id. Used everywhere else.
name	string	Display name including brand and pack size.
brand, category, subcategory	string / null	
category_ids	string[]	Category chain, root first.
description, image_url, has_image, image_version, images, updated_at	mixed	Present in real responses but not typed by the client.
unit, unit_quantity	string, number	For example kg and 0.75 for 750 g.
price_stats	object / null	min_price, max_price, avg_price, retailer_count, min_unit_price, last_computed. null when no retailer has a price.
retailers	string[]	Ids of retailers carrying the product.
retailer_prices	object[]	See below.
available_countries, is_international, private_label	mixed	private_label_retailer and its logo appear for store brands.
price_drop, price_drop_percentage, price_drop_from/to/start/end/category/title/company	mixed, nullable	Discount tracking data. Not modeled by the client.
gtin / barcode, barcode_aliases	string / string[]	Typed by the client but often absent.

retailer_prices item

Field	Type	Notes
retailer	string	Retailer id, matches /meta/retailers id.
retailer_display_name	string	The store's display name. Use this for the store.
retailer_name	string	The product's name as listed by that retailer (retailers may name the same product differently), not the store name. Can be an empty string, as in the competitors sample.
price	number	Current shelf price.
price_normalized	number	Price per unit (for example per kg).
is_discount, discount_percentage	bool, number / null	
last_updated	string	ISO datetime, no timezone.
country	string	ISO code, for example GR.

Paged list envelope (endpoints 1 and 2)

```
{
  "products": [ { ...product... } ],
  "total": 42, "page": 1, "page_size": 10,
  "total_pages": 5, "has_next": true, "has_prev": false,
  "query_time_ms": 12
}
```

Endpoint 1: POST /products/search

Status: **WORKING** (used in practice); response fields from client types

Used by tool(s): search_products

Search by text (title) or by a list of ids. This is what setup_products.py calls. Parameters go in a JSON body, not the URL.

Request body

Field	Type	Default	Notes
title	string	none	Search text. Only sent when non-empty.
ids	string[]	none	Look up specific product ids instead of text.
page	integer	1	Clamped to 1 to 10000.
page_size	integer	10	Clamped to 1 to 50.
sort_by	name price unit_price	name	
sort_order	asc desc	asc	

Example

```
POST /products/search
{ "title": "γάλα", "page": 1, "page_size": 10, "sort_by": "name", "sort_order": "asc" }
```

Response: the paged list envelope from Common objects, with product objects in products[]. The MCP tool additionally adds a source field and trims each product to a fixed subset of fields; calling the API directly returns the full product objects.

Endpoint 2: GET /products

Status: **FROM CLIENT CODE ONLY** (not tested against the live API)

Used by tool(s): list_products_by_category

Lists products in one category. Everything is passed as query parameters.

Param	Type	Default	Notes
category	string	required	Category id from /meta/categories/tree.
countries	string	GR	Fixed by the client.
page	integer	1	Clamped 1 to 10000.
page_size	integer	10	Clamped 1 to 50.
sort_by	name price unit_price	unit_price	Note the different default from search.
sort_order	asc desc	asc	
is_international	boolean	omitted	Only sent when set.

```
GET /products?category=<id>&countries=GR&page=1&page_size=10&sort_by=unit_price&sort_order=asc
```

Response: the paged list envelope from Common objects.

Endpoint 3: GET /products/{id}

Status: CONFIRMED with a real response sample

Used by tool(s): get_product, get_price_history

Full detail for one product including the price at every retailer. With include_history=true it also returns daily price history, the same data behind the price chart on the website.

Param	Type	Default	Notes
id (path)	string	required	Product id.
countries	string	GR	
include_tax	boolean	true	
sort_retailers	asc desc	omitted	Orders retailer_prices.
include_history	boolean	omitted	Adds the history block. Much larger response.
price_type	price unit_price	omitted	Which price the history tracks (the tool sends price).

```
GET /products/157cf7b3f6b440009c7cda657e96178b?countries=GR&include_tax=true&include_history=true
```

Response

A single product object (see Common objects), including fields the client never typed. With include_history=true a history block is added:

```
"history": {
  "retailers": ["synka", "galaxias", "mymarket", "kritikos", "..."],
  "date_range": { "start": "2026-07-18", "end": "2026-09-16" },
  "available_countries": ["GR"],
  "daily_prices": {
    "synka": [
      { "date": "2026-07-20", "country": "GR", "price": 1.68, "unit_price": 2.24 },
      { "date": "2026-07-21", "country": "GR", "price": 1.68, "unit_price": 2.24 }
    ],
    "galaxias": [ ... ]
  }
}
```

- daily_prices is a dict keyed by retailer id, each a list with one entry per day.
- There is no ready-made minimum. Flatten all price values and take min() yourself.
- The window is bounded by date_range (about two months in the sample), so a minimum computed from it is a recent low, not an all-time low.
- price_stats.min_price is only today's cheapest current price, not a historic figure.

Endpoint 4: GET /products/barcode/{barcode}

Status: FROM CLIENT CODE ONLY (not tested against the live API)

Used by tool(s): lookup_barcode

Same response as endpoint 3 but looked up by EAN/GTIN. Not tested live: the PosoKanei website does not display barcodes, so there was no barcode source to test with.

Param	Type	Default	Notes
barcode (path)	string	required	8, 12 or 13 digits. The client rejects anything else before sending.
countries	string	GR	
include_tax	boolean	true	

Endpoint 5: GET /products/{id}/competitors

Status: CONFIRMED with a real response sample

Used by tool(s): compare_competitors

Despite the name, competitors are other products, not other supermarkets. The endpoint returns products from the same category as the one requested, as sold by one chosen retailer.

Param	Type	Default	Notes
id (path)	string	required	Product id.
retailer	string	required in practice	Retailer id such as galaxias. Omitting it returns 400 Bad Request, although the client code marks it optional.
include_tax	boolean	true	

```
GET /products/37724eb931f84ebc8d809b1aaa204bdc/competitors?retailer=galaxias&include_tax=true
```

Response

```
{
  "id": "37724eb931f84ebc8d809b1aaa204bdc",
  "product_name": "ΛΟΥΜΙΑΔΗΣ ΠΑΠΑΓΑΛΟΣ Καφές Ελληνικός Σκούρος 490g",
  "category_id": "b2a17c2ad4235ea8574d60276381f3aa",
  "category_name": "Ελληνικός Καφές",
  "category_depth": 3,
  "total_competitors": 25,
  "competitors": [
    {
      "id": "2a1d38aa2d3f4968ab41a248fd03d63b",
      "name": "ΛΟΥΜΙΑΔΗΣ ΠΑΠΑΓΑΛΟΣ Καφές Ελληνικός Έκαστος Κοσμοπολίτης 96g",
      "private_label": false,
      "price_drop": false, "price_drop_percentage": null,
      "price_drop_from": null, "price_drop_to": null,
      "image_url": "https://api.posokanei.gov.gr/images/product/2a1d...",
      "category_ids": [...],
      "unit": "kg", "unit_quantity": 0.096,
      "retailer_prices": [
        {
          "retailer": "galaxias", "retailer_display_name": "Γαλαξίας",
          "retailer_name": "", "price": 2.48, "price_normalized": 25.83,
          "is_discount": false, "discount_percentage": null,
          "last_updated": "2026-09-25T00:00:00", "country": "GR" }
      ],
      "price_stats": { "min_price": 2.48, "max_price": 2.48, "avg_price": 2.48,
        "retailer_count": 1, "min_unit_price": 25.83, "last_computed": null }
    }
  ]
}
```

- Every competitor has the same field set. retailer_prices has 0 or 1 entries because of the retailer filter.
- When retailer_prices is empty (1 of 25 in the sample), price_stats is **null**, not an empty object.

Endpoint 6: GET /meta/categories/tree

Status: **CONFIRMED** with a real response sample

Used by tool(s): list_categories

Param	Type	Default	Notes
include_counts	boolean	true	Include product counts per category.
include_hidden	boolean	false	Include categories the website hides.

Depth and item limits are not API parameters. The MCP tool applies maxDepth (default 2) and maxItems (default 100) itself after downloading the whole tree.

Response

```
{
  "total_categories": 2682,
  "root_count": 19,
  "tree": [
    {
      "category_id": "cFsywHNrftQ6yeittltcSFQ4qbM14Q3F",
      "name": "Βρεφικά", "name_en": "Infants",
      "depth": 0, "hidden": false,
      "image_url": "https://api.posokanei.gov.gr/images/category/cFsyw...?v=1779894968",
      "vat_rates": null,
      "effective_vat_rates": {
        "GR": { "rate": 24.0, "source": "Standard rate", "source_id": null, "inherited": true }
      },
      "product_count": 0,
      "total_product_count": 267,
      "children": [ ...same shape, recursively... ]
    }
  ]
}
```

Field	Notes
category_id	Mostly 32-character hex, but 29 of 445 nodes use dashed UUIDs.
product_count / total_product_count	Direct products vs products including all descendants. Roots have product_count 0.
vat_rates	Usually null. When a category overrides VAT it is a flat map such as {GR: 6.0, BE: 6.0}.
effective_vat_rates	Per country {rate, source, source_id, inherited}. inherited false means this category sets it.
image_url	Absolute URL or null.

Observed in the sample

- Six roots: Βρεφικά (Infants, 267 products), Είδη για Κατοικίδια (Pet Shop, 229), Καθαριότητα (Cleaning, 1040), Ποτά (Drinks, 1429), Προσωπική Φροντίδα (Personal Care, 1537), Τρόφιμα (Food, 5218).
- 445 nodes in total, nesting to depth 5 (nodes per depth: 6, 37, 118, 227, 56, 1).
- The counters disagree with the tree: root_count says 19 and total_categories says 2682, but the tree contains 6 roots and 445 nodes. The counters likely cover a larger taxonomy than the tree returns.
- effective_vat_rates repeats about 11 countries on every node, which makes this response heavy.

Endpoint 7: GET /meta/retailers

Status: **CONFIRMED** with a real response sample

Used by tool(s): list_retailers

Param	Type	Default	Notes
countries	string	all	Tested with all (23 retailers, 11 countries). Filtering by GR was not tested.

Response

```
{
  "count": 23,
  "retailers": [
    { "id": "alphamega", "name": "Alphamega", "country": "CY",
      "logo_url": "/images/retailer/alphamega", "website": "https://www.alphamega.com.cy" },
    { "id": "ab_vasilopoulos", "name": "AB Βασιλόπουλος", "country": "GR",
      "logo_url": "/images/retailer/ab_vasilopoulos", "website": "https://www.ab.gr" }
  ]
}
```

Field	Present	Notes
id	23/23	Matches retailer values in retailer_prices.
name	23/23	Display name in the chain's own language.
country	23/23	ISO code.
logo_url	22/23	Relative path. Prepend the base URL. Missing for frisco.
website	21/23	null for edeka24 and synka.

Retailers returned

Country	Retailer ids
GR (11)	bazaar, lidl, market_in, mymarket, ab_vasilopoulos, galaxias, kritikos, masoutis, synka, sklavenitis, halkiadakis
Other (12)	alphamega (CY), auchan (FR), carrefour_es (ES), carrefour_it (IT), colruyt (BE), conad (IT), continente (PT), delhaize (BE), edeka24 (DE), freshful (RO), frisco (PL), ebag (BG)

The client types list many alias fields (display_name, retailer, retailer_display_name). None of them exist in the real response: only id, name, country, logo_url and website. Note that the price data uses retailer_display_name while this endpoint uses name for the store, and the two can differ in spelling (for example Synka in price data versus ΣΥΝ.ΚΑ here).

Endpoint 8: GET /meta/stats

Status: **CONFIRMED** with a real response sample

Used by tool(s): get_stats

No parameters.

```
{
  "total_products": 10420,
  "active_products": 9950,
  "retailers": ["masoutis", "sklavenitis", "galaxias", "...", "edeka24"],
  "retailer_count": 23,
  "products_on_discount": 4485,
  "timestamp": "2026-09-27T08:37:26.341014"
}
```

The response matches the client type exactly. retailers is the same set of ids as /meta/retailers.

Endpoint 9: GET /meta/categories

Status: EXISTS IN CLIENT CODE BUT NO TOOL USES IT (untested)

Used by tool(s): none

The client's listCategories() can call this flat variant when its tree option is off, but no MCP tool ever does. Its response shape is unknown.

Practical notes and gotchas

- **403 Forbidden:** send browser-like headers (see Overview). Plain script defaults are blocked.
- **400 on competitors:** the retailer parameter is required even though the client marks it optional.
- **Competitor means product:** endpoint 5 returns similar products, not other stores. Cross-store prices come from retailer_prices in endpoint 3.
- **price_stats can be null:** check for None before reading it.
- **Historic minimum:** compute it from history.daily_prices (endpoint 3, include_history=true). The window is about two months. price_stats.min_price is not historic.
- **retailer_name is the product name, not the store:** it holds the product's name as that retailer lists it (and is sometimes empty). Use retailer_display_name for the store.
- **logo_url is relative** in /meta/retailers. Image URLs in product data are absolute.
- **Page size is capped at 50** and page at 10000 by the client; the API may enforce its own limits.
- **Be gentle:** wait at least 250 ms between calls and cache results. History responses are large (thousands of lines for one product).
- **Client types are unreliable:** wherever a real response was inspected, it differed from or extended the posokanei-mcp TypeScript types. Treat untested rows in this document as provisional.

Minimal Python example

```
import requests

API = "https://api.posokanei.gov.gr"
HEADERS = {
    "accept": "application/json", "content-type": "application/json",
    "user-agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 "
        "(KHTML, like Gecko) Chrome/128.0.0.0 Safari/537.36",
    "origin": "https://posokanei.gov.gr", "referer": "https://posokanei.gov.gr/",
}

# 1. search
body = {"title": "ντομάτα πασάτα", "page": 1, "page_size": 5,
        "sort_by": "name", "sort_order": "asc"}
hits = requests.post(f"{API}/products/search", json=body, headers=HEADERS).json()["products"]

# 3. detail with history
pid = hits[0]["id"]
p = requests.get(f"{API}/products/{pid}", headers=HEADERS,
                 params={"countries": "GR", "include_tax": "true",
                         "include_history": "true"}).json()
prices = [d["price"] for series in p["history"]["daily_prices"].values() for d in series]
print(p["name"], "historic min:", min(prices))
```